

Marion Sebastian Nereveetil

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PROFESSIONAL SUMMARY

Computer Science undergraduate with strong interest in data-driven systems, automation, machine learning, and applied AI. Hands-on experience building real-time analytical systems, computer vision applications, deep learning models, and optimization-focused solutions. Strong foundation in Python, data structures, and problem solving with focus on building reliable and efficient systems.

TECHNICAL SKILLS

- **Programming Languages:** Python, C, C++, Java
- **Data & AI:** Data Analysis, Machine Learning, Deep Learning, Time-Series Forecasting, Computer Vision
- **Computer Vision:** OpenCV, YOLO, Facial Landmark Detection
- **Core Concepts:** Data Structures, Algorithms, Object-Oriented Programming, DBMS, Operating Systems
- **Tools:** Git, GitHub, Version Control

PROJECTS

Air Quality Forecasting using Deep Learning

Python, TensorFlow

- Developed a time-series forecasting system using historical air-quality data from India (2010–2023).
- Built a deep learning pipeline for preprocessing, feature engineering, normalization, and pollutant trend prediction.
- Integrated live air-quality data using the Open-Meteo API for real-time forecasting and comparative analysis.
- Visualized pollutant trends, model predictions, and time-series patterns for performance evaluation.

Driver Drowsiness Detection System

Python, OpenCV

- Developed a real-time monitoring system to detect driver fatigue using Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR).
- Implemented facial landmark detection on live video streams to identify alertness patterns.
- Designed automated alert triggers to improve operational safety and response time.

Lane and Vehicle Detection using YOLO

Python, YOLO

- Built a computer vision pipeline for detecting vehicles and road lanes from video input.
- Applied YOLO-based object detection for real-time identification and localization.
- Optimized inference flow to balance detection accuracy and processing speed.

Pomodoro Timer Web Application

HTML, CSS, JavaScript

- Developed a productivity-focused web application with configurable work and break cycles.
- Implemented client-side logic for time tracking and user interaction.
- Focused on clean UI design and reliable application behavior.

EDUCATION

Bachelor of Technology in Computer Science

Vellore Institute of Technology (VIT)

2023 – Present

- CGPA: 7.93 / 10

- Relevant Coursework: Data Structures, DBMS, Operating Systems, Object-Oriented Programming, Computer Networks